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CASE REPORT

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Advanced-stage colorectal cancer in the context of coronavirus disease-19, highlighting socioeconomic disparities and outcomes: a case report

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Abstract

Background This is a case report about how pre-coronavirus disease-19 pandemic health disparities in African Americans with colorectal cancer were compounded and exacerbated by the pandemic's associated delays in screening.

Case presentation A 69-year-old African American male patient presented with a primary concern of hematochezia for 1 week. His history revealed multiple risk factors for colorectal cancer, including obesity, diabetes mellitus, and a family history of colorectal cancer in his grandfather. Of note, his father had prostate cancer. On colonoscopy, the patient was found to have a 3-cm fungating, ulcerated, partially obstructing mass in the sigmoid colon, histologically confirmed as well-differentiated adenocarcinoma with mismatch repair proficiency. Imaging studies demonstrated a 2.4-cm hepatic lesion consistent with metastatic disease. Despite a history of benign polyps, removed 10 years earlier, the lack of follow-up colonoscopy led to a delayed diagnosis. The patient was diagnosed with stage IV colorectal adenocarcinoma with hepatic metastases. Initial treatment involved systemic chemotherapy with the folinic acid, fluorouracil, and oxaliplatin regimen to address micrometastatic disease. A laparoscopic hepatectomy and sigmoid colectomy with primary anastomosis was subsequently performed. After surgery, the patient underwent additional chemotherapy with folinic acid, fluorouracil, and oxaliplatin and bevacizumab for 3 months. Despite this, surveillance imaging revealed disease progression within 4 months, including new lung nodules and remnant liver disease. At this stage, therapy was escalated to folinic acid, fluorouracil, and irinotecan and cetuximab owing to the tumor's microsatellite stability and *K-RAS/BRAF* wild-type status. Unfortunately, the patient's disease progression reflects the aggressive nature of advanced-stage colorectal cancer and highlights the challenges of managing such cases when diagnosis is delayed.

Conclusion This case underscores the critical role of timely colorectal cancer screening and follow-up in preventing late-stage presentations, particularly among African Americans, who already face a disproportionate burden of colorectal cancer incidence and mortality. Socioeconomic barriers, compounded by disruptions during the coronavirus disease-19 pandemic, were key contributors to the patient's delayed diagnosis and subsequent poor outcome. This highlights the need for comprehensive approaches to improve healthcare access, including patient navigation programs, policy changes to expand insurance coverage, and community outreach initiatives to increase awareness about the importance of screening. In addition, early and consistent communication of family cancer history, even

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involving second-degree relatives, may impact surveillance schedules. This case is also of relevance to the general population as the coronavirus disease-19 pandemic-associated colorectal cancer screening delays affected all populations and as such, preventative strategies need to be put in place to address missed screenings and diagnoses.

Keywords African Americans, Colorectal cancer, Disparity, COVID-19, Case report

Introduction

Colorectal cancer (CRC) is the third most common cancer and the second leading cause of cancer-related deaths globally and in the USA [1]. The American Cancer Society estimates over 150,000 new cases of cancer and approximately 52,000 deaths in the USA for 2024. African Americans (AA) have higher CRC incidence (41.9 versus 37.0 per 100,000) and mortality (16.8 versus 12.9 per 100,000) compared with white individuals [2]. Despite an overall decline in CRC rates owing to improved screening, African Americans have lower screening rates when compared with non-Hispanic white people (55.6% versus 59.2%). The coronavirus disease-19 (COVID-19) pandemic led to a 90% drop in colonoscopies, with an estimated 18,800 missed CRC diagnoses and 1.7 million missed colonoscopies, potentially resulting in over 4,500 excess CRC deaths over the next decade [3]. The COVID-19 pandemic was associated with an increased diagnosis of advanced-stage CRC, evidenced by an increase in metastatic cases [4]. Insurance disparities, especially affecting AA populations, contribute to lower screening rates and canceled colonoscopies. Pronounced staging disparities disproportionately affected AA patients and the uninsured, highlighting the urgent need for targeted interventions in these vulnerable populations [5, 6].

Herein, we present a case report of a 69-year-old African American male with significant medical history of diabetes and hypertension who presented with gastrointestinal (GI) bleeding and was ultimately diagnosed with stage IV CRC with liver metastasis. The patient's disease progression was notably influenced by socioeconomic barriers, including a lack of insurance that delayed timely surveillance colonoscopy, compounded by healthcare access delays during the COVID-19 pandemic. Despite initial successful treatment with systemic chemotherapy and surgical intervention, the patient experienced CRC recurrence within 4 months.

Case presentation

A 69-year-old African American male with a medical history of obesity, diabetes mellitus, hypertension, smoking, and drug abuse presented to the clinic complaining of hematochezia for 1 week. The patient's family history was remarkable; his grandfather had CRC and his father has prostate cancer. The patient denies abdominal pain, abdominal distention, diet intolerance, nausea, vomiting, or weight loss. According to the patient, he underwent screening colonoscopy in another institution about 10 years previously in which some benign polyps were found (report was not available) and he was supposed to follow up with a repeat colonoscopy 5 years later but he could not do so because he lost his health insurance coverage. For evaluation of GI bleeding, colonoscopy was performed, and findings included a 3-cm fungating, ulcerated, partially obstructing mass in the sigmoid colon, histopathology revealed a well-differentiated adenocarcinoma with mismatch repair (MMR) proficient (Fig. 1). Additional findings included a 6-mm sessile polyp in the transverse colon (tubular adenoma), a 4-mm sessile polyp in the descending colon (hyperplastic), and a 6-mm sessile polyp in the sigmoid colon (hyperplastic).

Staging computed tomography (CT) scans of the abdomen and pelvis revealed a proximal sigmoid colon intraluminal fungating lesion measuring 3.1 × 2.5 × 3.2 cm and a 2.4-cm hepatic lesion suggestive of metastatic disease; no lesions were identified in lungs (Fig. 2). A magnetic resonance imaging (MRI) scan of the pelvis confirmed the secondary liver metastasis. Laboratory evaluation showed a carcinoembryonic

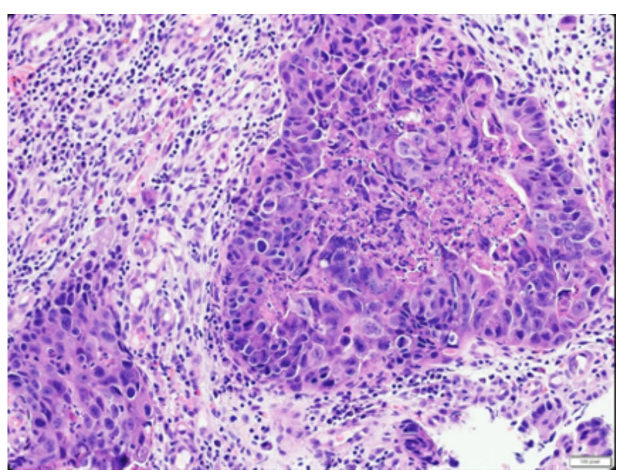

Fig. 1 Hematoxylin and eosin-stained histology image (20×) of poorly differentiated sigmoid adenocarcinoma of the colon; the tumor is characterized by irregular, atypical cells with a high nucleus-to-cytoplasm ratio, a lack of distinct glandular architecture, infiltrative growth patterns, and desmoplastic stroma

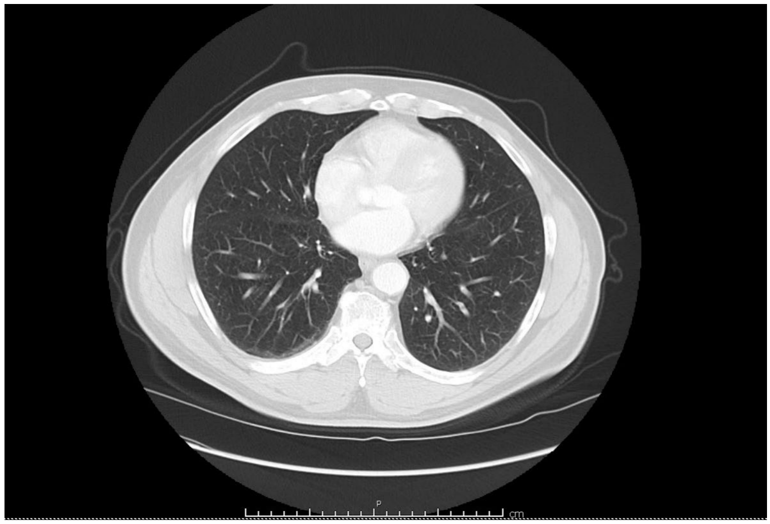

Fig. 2 No metastatic lesion identified in lungs during staging computed tomography scan

antigen (CEA) tumor marker level of 3.8 ng/mL, hemoglobin at 14.2 g/dl, and hemoglobin A1c at 6.9 g/dl.

Follow-up and outcome

The case was a radical surgery, given the stage IV synchronous metastatic disease of his CRC, and the consensus was to proceed with systemic chemotherapy first to address any micrometastatic disease and was followed by robotic-assisted hepatectomy and robotic-assisted lap sigmoid colectomy. Systemic chemotherapy was done using the leucovorin calcium (folinic acid), fluorouracil, and oxaliplatin (FOLFOX) regimen to address any micrometastatic disease. After completion of four cycles, the patient underwent laparoscopy hepatectomy and sigmoid colectomy with primary anastomosis. The following 3 months, he completed four more cycles of modified FOLFOX with bevacizumab (Avastin). However, the surveillance studies demonstrated the progression of the disease with new multiple nodules in the lungs and remnant liver (Fig. 3). Given the tumor was microsatellite stable (MSS) and *K-RAS*/*BRAF* wild-type, the patient was treated with six cycles of leucovorin calcium (folinic acid), fluorouracil, irinotecan hydrochloride (FOLFIRI), and cetuximab. The timeline and clinical course of the patient is depicted in Fig. 4.

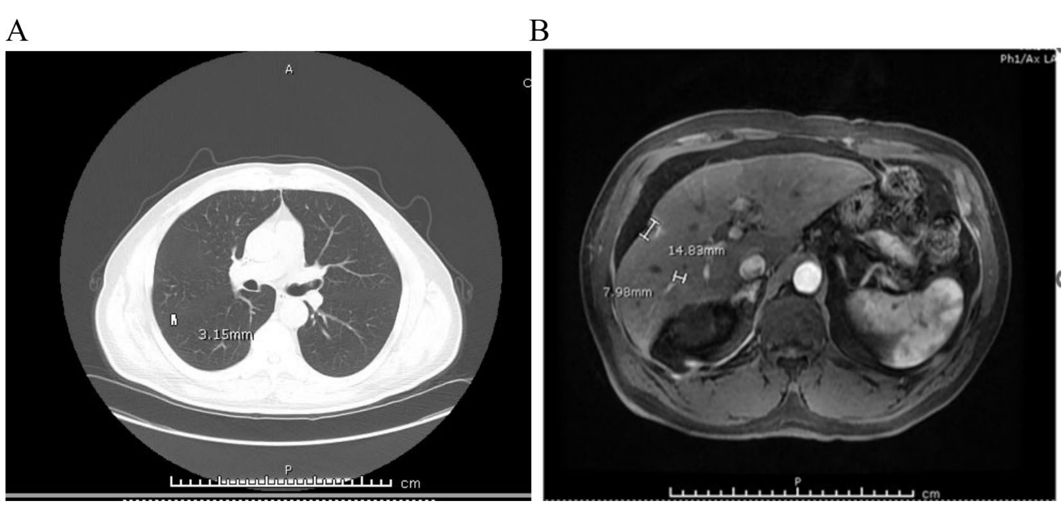

Fig. 3 Right lung nodules and remnant liver disease in surveillance computed tomography scan 3 months post chemotherapy (A). The computed tomography scan revealed potential disease progression, with the presence of tiny lung nodules, suggesting the development of metastatic disease (B)

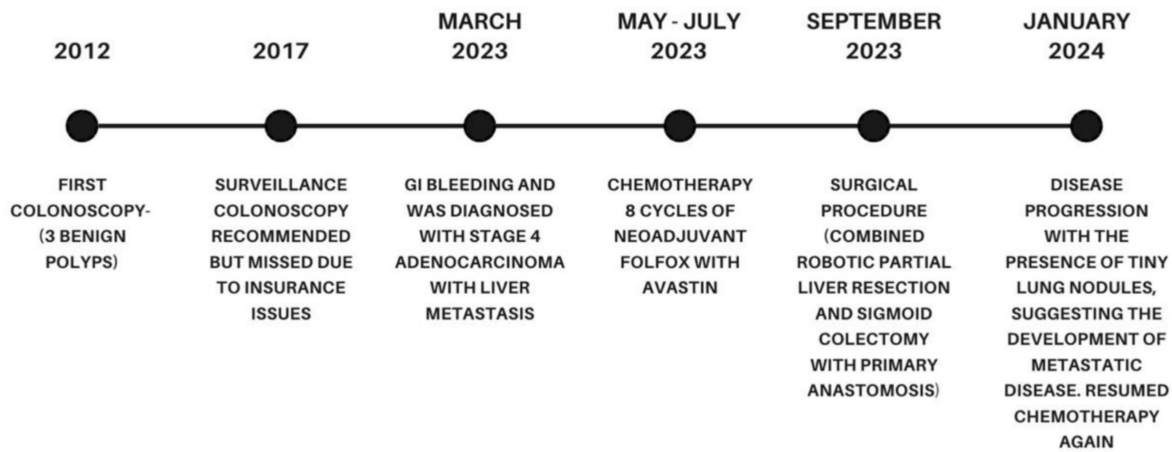

Fig. 4 Timeline and clinical course

Discussion and conclusion

This case highlights the significant impact of lack of appropriate insurance/healthcare access issues accentuated during the COVID-19 pandemic with worsening of CRC burden among vulnerable populations. Despite a history of benign polyps removed in 2012, the patient missed a critical surveillance colonoscopy in 2017 owing to cancellation caused by lack of health insurance coverage. The disruption in care worsened during the COVID-19 pandemic, as only emergency procedures were prioritized and routine screenings and surveillance colonoscopies for precancerous polyps were suspended at that time, leading to further missed cancer prevention opportunities. By the time symptoms appeared in 2023, the patient's cancer had progressed to stage IV with metastatic disease. Despite appropriate management, the patient had cancer recurrence within 4 months. It is conceivable that accessible and timely postpolypectomy surveillance could have detected the adenocarcinoma earlier, potentially improving outcomes. This case underscores the critical need for accessible preventive screening in at-risk populations, as early detection could have significantly improved the patient's prognosis, management, and treatment outcomes.

Unfortunately, African Americans with CRC are often diagnosed at younger ages, have advanced stages at presentation, higher overall mortality, and worse 5-year overall survival compared with white individuals [2, 7]. A multifaceted approach is needed to address the impact of missed follow-ups and disparities in CRC among AA. Enhancing patient education and awareness through culturally sensitive materials and community outreach is crucial, emphasizing the importance of routine screenings, and timely follow-ups. Implementing patient navigation programs can significantly assist patients in overcoming barriers such as transportation, financial challenges, and healthcare system navigation and access.

Missed follow-ups in patients with high risks of CRC have significant implications for both the patient and the healthcare system. For patients, missed follow-up appointments can lead to delays in detecting cancer early, reducing progression, and preventing recurrence, which generally result in worsened outcomes, reduced survival rates, and diminished quality of life. These delays can also increase the physical and emotional burden on patients and their families, as they may face more aggressive treatments and uncertain prognoses. From a healthcare cost perspective, the lack of timely follow-up often results in the need for more intensive medical interventions, increased hospitalizations, and more extended hospital stays, all of which contribute to higher healthcare expenditures. In addition, poor follow-up can lead to unnecessary diagnostic tests and procedures, further escalating costs. Overall, the loss of follow-up in patients with CRC is a critical issue that exacerbates health disparities, increases patient morbidity and mortality, and places a significant financial strain on the healthcare system.

The COVID-19 pandemic has exacerbated disparities in CRC outcomes and follow-up care, particularly among African American patients, while also disproportionately affecting these communities with higher infection and mortality rates, thereby increasing the strain on an already vulnerable population [8–10]. During the pandemic, many routine screenings, elective procedures, and follow-up appointments were delayed or canceled owing to restrictions on nonemergent medical services and patient concerns regarding potential exposure to the virus. This disruption in care led to significant delays in diagnosis and treatment for many patients, particularly those already facing barriers to healthcare access. The combination of healthcare system disruptions and increased health risks has widened the existing disparities in colorectal cancer outcomes, leading to even more delayed diagnoses, advanced disease stages at presentation, and higher mortality rates. This has also increased the financial burden on the healthcare system and patients, as treating more advanced cancer requires more intensive and costly interventions. Addressing these disparities requires renewed efforts to ensure continuity of care and targeted support for the most affected communities.

Leveraging technology, such as electronic health records with alert systems and telemedicine options, can improve communication and facilitate more effortless follow-up for patients who may face difficulties attending in-person appointments. Furthermore, addressing socioeconomic and systemic barriers is essential. This includes advocating for policies that reduce healthcare access barriers, such as expanding insurance coverage, patient advocacy, and improving transportation options. Several of the known barriers and challenges to adequate screening have been further exacerbated during the COVID-19 pandemic as a result of delayed nonemergent procedures, leading to missed follow-up, as is the case with our patient who lost his health insurance coverage initially and experienced subsequent delays in getting a follow-up colonoscopy for his precancerous colon polyps.

Participating in quality improvement initiatives is vital to reducing missed follow-ups. Analyzing data on follow-up adherence and outcomes can identify gaps and implement targeted interventions. Furthermore, promoting research and policy advocacy focused on understanding the root causes of disparities and advocating for structural changes in healthcare will help create a more equitable system to eradicate CRC-associated disparities.

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Author contributions
H.A. and H.B. contributed to the conception and design of the study. Material preparation and data collection was performed by M.D., R.Z., M.R., H.B., and H.A. Funding acquisition by performed by H.A. and H.B. Review and edits done by A.L.. All authors read and approved the final manuscript. H.A. designed the study; H.A. and H.B. wrote the manuscript; S.G. and M.D. reviewed and edited the paper; and C.N., R.Z., and A.O.L. collected and analyzed the clinical data.

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Data availability
The de-identified data are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The human subject for the case study was approved by Howard University institutional review board (IRB)#12-CMED-79.

Consent for publication
Written informed consent was obtained from the patient for publication of this case report and any accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal.

Competing interests
The authors declare that there are no conflicts of interest regarding the publication.

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